LgdDecompiler V 0.1.0

🌍 **Language** / 语言

- US English Version
- CN 中文版

US English Version

lgd Manual

If you are interested in learning more about lgd files, please visit

lgd Manual MD

Or

lgd Manual PDF

How to Use

Simply drag and drop a directory containing `.lgd` files, or a single `.lgd` file, onto the corresponding `.bat` script.

You have the option to keep the intermediate files generated during the analysis process, or clear them to strictly output only the final `.LGC` files.

During execution, please ignore any yellow `[Warning]` logs. As long as there are no red `[Error]` logs, the decompiled scripts should be syntactically sound.

> ***Note:*** *It is not recommended to use this tool on versions containing* `default.lgd`*. It will cause severe lag and will likely freeze the program.*

Project Structure

```
LgdDecompiler/
├── main.py                    # Program entry point
├── config.py                  # Global configuration
├── logger.py                  # Logging module
|
├── common/                    # LGD file definition modules
|   ├── binary_utils.py        # Utils: XOR decryption and string formatting
|   ├── definitions.py         # Part 1 & Part 2 table definitions
|   ├── opcodes.py             # VM opcode mapping and instruction definitions
|   └── structures.py          # Overall LGD structural blueprints
|
├── core/                      # Initial analysis and conversion modules
|   ├── analyzer.py            # Further analysis of wrapped objects
|   ├── context.py             # State management for the decompilation context
|   ├── parsers.py             # Parses raw data and wraps into objects
|   └── pipeline.py            # Main pipeline scheduler
|
├── generate_intermediate/     # Intermediate product generation
|   ├── lgd_decryptor.py       # Decrypts the raw .lgd binary file
|   ├── renderer_asm.py        # Converts bytecode into readable .asm for analysis
|   ├── renderer_c.py          # Experimental analyzer for P1 & P2 (for manual review)
|   └── exporter_csv.py        # Exports string/extern tables to CSV format
|
├── generate_LGC/              # Core LGC code generator
|   ├── asm_parser.py          # ASM Parser: Reads .asm and converts to structured objects
```

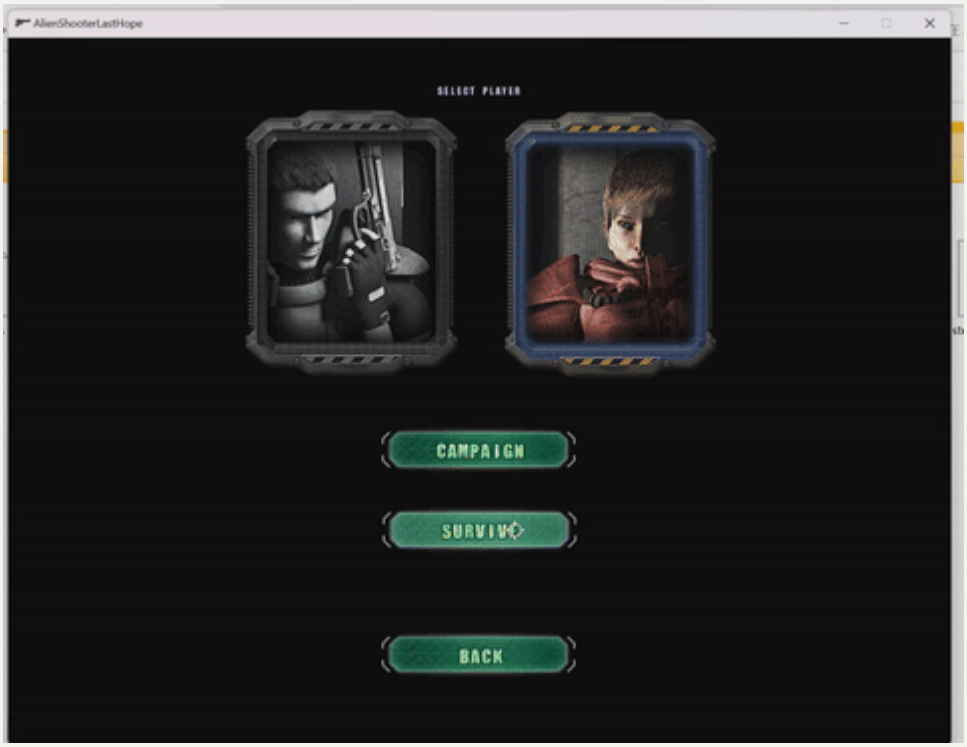
```
|   ├── asm_structs.py          # Data structures for assembly instructions and methods
|   ├── cfg_builder.py          # CFG Builder: Splits Basic Blocks and jump edges
|   ├── cfg_structs.py          # Data structures for CFG basic blocks
|   ├── ast_builder.py          # AST Builder: Derives expressions via symbolic execution
|   ├── ast_structs.py          # AST Nodes (Variables, constants, binary ops, etc.)
|   ├── flow_structurer.py      # Flow Structurer: Identifies If/While/For via graph theory
|   ├── lgc_context.py          # LGC specific context (local variables, type inference)
|   ├── lgc_header_manager.py   # Header Manager: Handles globals, externs, and arrays
|   ├── lgc_optimizer.py        # Optimizer: Simplifies AST structures and redundant expressions
|   └── lgc_generator.py        # Code Generator: Traverses AST to output final scripts
|
├── LGC_refiner/                # Post-processing code optimizer (Planned)
|   └── lgc_refiner.py
|
└── test/                       # Test suites
```

Known Bugs

Since this is an early v0.1.0 release, there are quite a few bugs. Currently, the decompiler barely reaches the point where the script can compile and load into the game.

1. Survival Mode Inevitable Crash

- **Scenario 1:** Crashes immediately upon clicking the start button.



- **Scenario 2:** Enters the game, but crashes immediately upon pressing ESC.

- **Scenario 3:** Enters the game, successfully returns to the menu via ESC, but crashes shortly after.

> ⚠️ ***Note:*** *After the crash, the program process will hang in the background and must be terminated manually via Task Manager.*

后台进程 (92)		
AcroTray (32 位)	0%	0.3 MB
Adobe IPC Broker (32 位)	0%	0.3 MB
AlienShooter Application (3...	8.6%	49.1 MB
AlienShooter Application (3...	9.6%	50.9 MB
AMD External Events Client ...	0%	0.9 MB
AMD External Events Servic...	0%	0 MB
AMD Software (4)	0%	8.3 MB
Antimalware Core Service	0%	4.1 MB
Antimalware Service Execut...	0%	92.3 MB
Application Frame Host	0%	0.8 MB

2. Crash After Mission Completion

This is a random event, possibly related to runtime duration or whether screen recording is active. It might crash immediately after clicking the clear-stage button, shortly after entering the next level, or after powering up and returning to the main menu. The first scenario is the most frequent.

Crash Logs:

```
# Crash immediately after clicking the button
[E] 16:10:01.399 - [thread_2] Unexpected action: 120,  nVid = 0 in 'int __thiscall SPRITE::Action(int,int,int,int)' <H:\repository\sources\sprite.cpp> at 2872
[E] 16:10:01.470 - [thread_2] SCRIPT Can't find variable "Action589943" in 'int __thiscall SCRIPT::GetVariableActionInt(unsigned int)' <H:\repository\sources\script.cpp> at 318
[E] 16:10:01.470 - [thread_2] SCRIPT Can't find variable "Action589943" for SetVariableActionInt in 'void __thiscall SCRIPT::SetVariableActionInt(unsigned int,int)' <H:\repository\sources\script.cpp> at 334
[E] 16:10:01.470 - [thread_2] Reference count non zero after delete -1,  nvid = 504 in '__thiscall SPRITE::~SPRITE(void)' <H:\repository\sources\sprite.cpp> at 463

# Crash after entering the next level
[I] 17:02:35.155 - [thread_2] Start to load map "maps\shop.map"
...
[E] 17:02:35.161 - [thread_2] Can't load menu ini: maps\shop.ini in 'void __thiscall MENU::fillMorphingList(const class STRING &)' <H:\repository\sources\menu.cpp> at 456
```

3. Monster Indicator Arrow Logic Error

The arrow cannot point to monsters dynamically. It currently spawns fixed on the ground and disappears only after all monsters are cleared. *Fix priority:* Moderate. Should be patchable by porting the logic from the *Lost City* version.

4. Lingering "Loading" Text on Maps

A "Loading" string occasionally gets stuck at fixed coordinates on the map. Suspected to be present in every level (confirmed on the first two levels).

CN 中文版

lgd文档

如果你有兴趣了解更多有关lgd文件的信息,请访问

lgd说明手册 MD

或

lgd说明手册 PDF

如何使用

将含有 `.lgd` 的文件夹路径或单个 `.lgd` 文件直接拖拽到对应的 `.bat` 脚本上即可执行。

你可以选择保留分析过程中的中间产物，或者清除它们只保留最终生成的 `.LGC` 源码文件。

在运行过程中出现的黄色 `[警告]` 日志可以忽略，只要不出现红色的 `[报错]`，生成的脚本在语法上至少是完整的。

> 注意： 极不建议用于处理包含 `default.lgd` 的版本，这会导致极其严重的卡顿甚至程序卡死。

项目结构

```
LgdDecompiler/
├── main.py                    # 程序入口点
├── config.py                  # 全局配置
├── logger.py                  # 日志模块
|
├── common/                    # lgd文件定义模块
|   ├── binary_utils.py        # 辅助工具：XOR解密算法和格式化字符串
|   ├── definitions.py         # LGD P1与P2表格中的基础定义
|   ├── opcodes.py             # 虚拟机操作码映射表与指令定义
|   └── structures.py          # LGD 总体结构蓝图
|
├── core/                      # 初步分析与转化模块
|   ├── analyzer.py            # 对封装对象进行进一步分析
|   ├── context.py             # 反编译对象上下文状态管理
|   ├── parsers.py             # 解析二进制并封装为对象
|   └── pipeline.py            # 总体任务调度
|
├── generate_intermediate/     # 中间产物生成
|   ├── lgd_decryptor.py       # 解密原始的 .lgd 二进制文件
|   ├── renderer_asm.py        # 将字节码转换为可读的 .asm 汇编代码，供后续分析使用
|   ├── renderer_c.py          # 对P1、P2的实验性早期分析版本，用于人工走读校验
|   └── exporter_csv.py        # 将字符串表、外部函数表等导出为 CSV
|
├── generate_LGC/              # LGC源码生成器核心
|   ├── asm_parser.py          # 汇编解析器：读取 .asm 文本并转换回结构化对象
|   ├── asm_structs.py         # 定义汇编指令和方法的数据结构
|   ├── cfg_builder.py         # 控制流图（CFG）构建器：切分基本块与跳转边
|   ├── cfg_structs.py         # 定义 CFG 基本块的数据结构
|   ├── ast_builder.py         # 抽象语法树（AST）构建器：通过符号执行推导表达式与堆栈状态
```

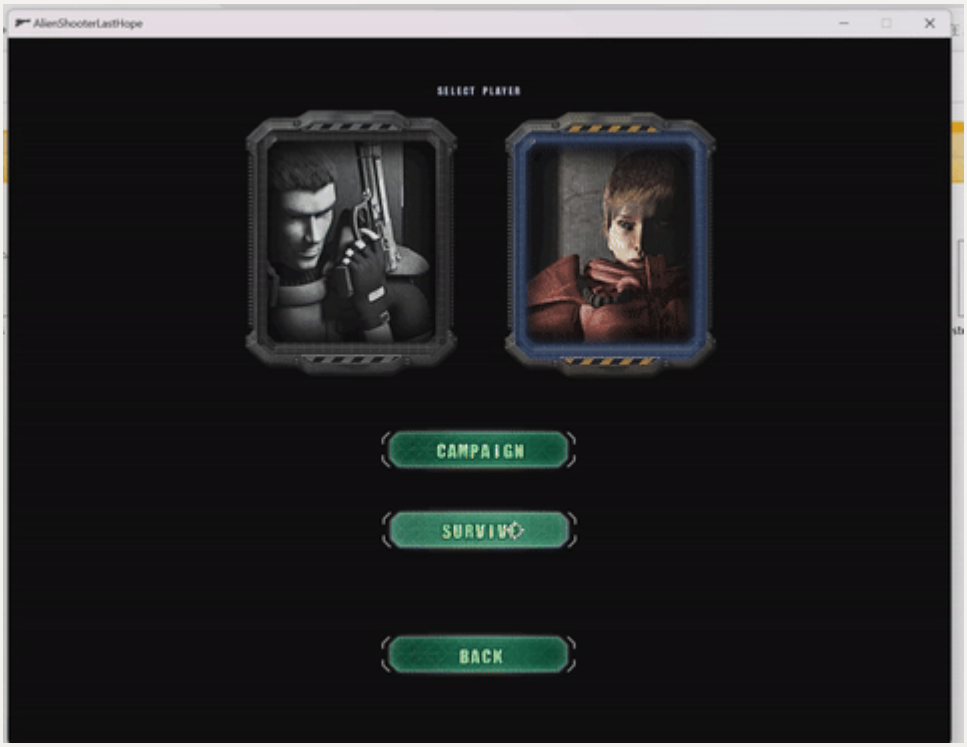
```
|   ├── ast_structs.py          # 定义 AST 节点 (变量、常量、二元运算、函数调用等)
|   ├── flow_structurer.py      # 控制流结构化器：利用图论和支配树识别还原 If/While/For 结构
|   ├── lgc_context.py          # LGC 代码生成的特定上下文（局部变量名、类型推导等）
|   ├── lgc_header_manager.py   # 头部管理器：处理全局变量、外部引用 (Externs) 和数组声明
|   ├── lgc_optimizer.py        # LGC 代码优化器：优化 AST 结构，简化冗余表达式
|   └── lgc_generator.py        # LGC 代码生成器：遍历结构化的 AST 树，输出最终脚本代码
|
├── LGC_refiner/                # 代码后期优化模块（开发中）
|   └── lgc_refiner.py
|
└── test/                       # 测试套件
```

已知 BUG

由于目前还是早期 0.1.0 版本，Bug 还比较多，目前仅能堪堪做到编译通过并且能挂载进游戏的地步。

1. 生存模式必定闪退

- 情况 **1**： 点击按钮后立刻崩溃。



- 情况 **2**： 进入游戏后，点击 ESC 立刻崩溃。

- 情况 **3**： 进入后点击 ESC 成功返回菜单，但停留一段时间后闪退。

> ⚠️ 注意： 闪退后程序会在后台幽灵挂载，需要打开任务管理器手动终止。

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Application Frame Host	0%	0.8 MB

2. 点击完成任务后闪退

这是一个概率事件，可能和游戏运行时长有关，也有可能和是否开启了录屏有关。 有可能在点击通关按钮时立刻闪退，也可能在进入下一关一段时间后闪退，或者在升级加点后点击 main menu 时闪退。其中第一种情况发生概率最高。

崩溃日志参考：

```
# 点击按钮后立刻报错
[E] 16:10:01.399 - [thread_2] Unexpected action: 120,  nVid = 0 in 'int __thiscall SPRITE::Action(int,int,int,int)' <H:\repository\sources\sprite.cpp> at 2872
[E] 16:10:01.470 - [thread_2] SCRIPT Can't find variable "Action589943" in 'int __thiscall SCRIPT::GetVariableActionInt(unsigned int)' <H:\repository\sources\script.cpp> at 318
[E] 16:10:01.470 - [thread_2] SCRIPT Can't find variable "Action589943" for SetVariableActionInt in 'void __thiscall SCRIPT::SetVariableActionInt(unsigned int,int)' <H:\repository\sources\script.cpp> at 334
[E] 16:10:01.470 - [thread_2] Reference count non zero after delete -1,  nvid = 504 in '__thiscall SPRITE::~SPRITE(void)' <H:\repository\sources\sprite.cpp> at 463

# 进入下一关后报错
[I] 17:02:35.155 - [thread_2] Start to load map "maps\shop.map" ...
[E] 17:02:35.161 - [thread_2] Can't load menu ini: maps\shop.ini in 'void __thiscall MENU::fillMorphingList(const class STRING &)' <H:\repository\sources\menu.cpp> at 456
```

3. 指向怪物的箭头逻辑有误

目前的脚本无法动态指向怪物。它只会在地上创建一个固定的箭头，并在全部杀完怪后才消失。 修复思路： 难度不大，后续可以从 *Lost City* 版本的源码中抄一份逻辑过来打补丁。

4. 地图上会残留 Loading 字样

疑似被硬编码打在了某个地图的固定坐标上，可能会在每一个地图中都有出现（只仔细验证了前两关的地图，均存在此问题，后续关卡待确认）。